

Cart Abandonment Pattern Analysis

SQL Analysis Summary — DuckDB on February Cart Events Dataset

Dataset scope: 2,656,529 cart-event rows and 1,660,109 cart sessions. The CSV contains cart events only; purchase outcome is represented by the boolean **purchased** field at session level.

Important: The findings below describe associations in the dataset. They do not prove that changing one factor will cause conversion to increase.

1. What is the overall purchase and cart abandonment rate?

Metric	Result
Total sessions	1,660,109
Purchased sessions	700,091
Abandoned sessions	960,018
Purchase / conversion rate	42.17%
Abandonment rate	57.83%

Finding: 57.83% of cart sessions did not result in a purchase, making abandonment the main baseline problem identified by the analysis.

Actionable recommendation: Use the 57.83% abandonment rate as the baseline KPI. Segment the abandoned sessions by engagement, category, brand and timing to identify where improvement opportunities are concentrated.

2. Does the number of distinct products affect purchase likelihood?

Outcome	Avg. distinct products/session
Abandoned	1.09
Purchased	1.21

Finding: Purchased sessions contained 0.12 more distinct products on average, about 11% higher than abandoned sessions.

Actionable recommendation: Investigate whether product discovery, cross-sell or recommendation features are associated with deeper carts. Treat this as a hypothesis rather than assuming that adding products causes conversion.

3. How does purchase behavior change with the number of distinct products?

Product band	Sessions	Purchase rate
1 product	1,484,011	39.84%
2 products	139,468	60.18%
3 products	25,332	67.09%
4+ products	11,299	69.65%

Finding: Purchase rate rises sharply from one product to two products, then the improvement becomes smaller. The 4+ group has the highest rate but much lower volume.

Actionable recommendation: Examine ways to help single-product sessions discover relevant additional products. Measure recommendation and cross-sell experiments separately rather than assuming more products will directly cause conversion.

4. Does product price relate to purchase behavior?

Price band	Sessions	Purchase rate
■0–■100	542,482	35.03%
■100–■250	577,423	48.03%
■250–■500	294,490	46.72%
■500+	316,126	46.01%

Finding: The ■0–■100 group has the lowest purchase rate. Above ■100, purchase rates are relatively similar.

Actionable recommendation: Investigate low-price cart sessions for differences in product mix, intent and browsing behavior. Do not assume that increasing or decreasing prices will improve conversion based on this analysis alone.

Note: Price here refers to carted product price, not total cart value.

5. Does session duration relate to purchase behavior?

Duration	Sessions	Purchase rate
0 minutes	1,292,378	36.08%
1–2 minutes	171,672	58.21%
3–5 minutes	91,104	68.99%
6–10 minutes	52,204	70.12%
10+ minutes	52,752	65.13%

Finding: Purchase rate rises strongly from zero-minute sessions to 6–10 minutes, then declines somewhat for 10+ minute sessions. The median session duration is 0 minutes.

Actionable recommendation: Focus on understanding why very short sessions abandon. Review friction points and engagement flows, while avoiding tactics designed simply to keep users on the site longer.

6. Does the number of cart events relate to purchase behavior?

Cart events	Sessions	Purchase rate
1 event	1,143,622	35.38%
2 events	308,456	53.49%
3 events	107,012	61.33%
4+ events	101,020	64.17%

Finding: Purchase rate increases as cart activity increases. The largest jump is from one to two cart events: +18.11 percentage points.

Actionable recommendation: Analyze what users do between the first and subsequent cart events. Product detail views, cart edits, recommendation interactions and checkout progression would be useful follow-up metrics if available.

7. Which product categories have higher or lower purchase rates?

Category	Sessions	Purchase rate
construction.tools.light	657,112	52.36%
sport.bicycle	118,266	38.71%
electronics.clocks	63,778	38.94%

Category	Sessions	Purchase rate
apparel.shoes	56,099	36.51%
appliances.personal.massager	51,696	45.78%
appliances.kitchen.coffee_grinder	13,856	60.62%
apparel.scarf	12,375	22.57%
sport.trainer	14,166	31.79%

Finding: Category purchase rates vary substantially. Among categories with at least 10,000 sessions, observed rates ranged from 22.57% to 60.62%.

Actionable recommendation: Prioritize category-level investigation using both volume and conversion rate. Review low-converting, high-volume categories first because they combine meaningful traffic with weaker purchase outcomes.

8. Does brand relate to purchase behavior?

Brand	Sessions	Purchase rate
apple	353,624	45.08%
samsung	305,404	54.73%
xiaomi	154,307	44.80%
huawei	47,575	54.15%
oppo	19,502	59.20%
sony	18,576	39.01%
bosch	13,825	37.58%
philips	10,032	35.80%

Finding: Brand-level purchase rates vary meaningfully among brands with at least 10,000 sessions, ranging from 35.80% to 59.20% in the analyzed group.

Actionable recommendation: Use brand-level metrics to identify combinations of high traffic and lower conversion. Investigate product mix and category differences before attributing performance differences to the brand itself.

9. Does purchase behavior vary by time of day?

Observed hour	Purchase rate
1 AM	33.63%
6–9 AM	43.90–44.05%
3–5 PM	38.29–38.85%
10 PM	45.84%

Finding: Purchase rate varies by hour, but the differences are moderate compared with the stronger engagement-related patterns. Some overnight hours also have lower session volumes.

Actionable recommendation: Use time-of-day as a supporting segmentation variable. Avoid making operational decisions from individual low-volume hours without additional evidence.

10. Does purchase behavior vary by day of the week?

Day	Sessions	Purchase rate
Sunday	258,210	41.95%
Monday	229,450	41.58%
Tuesday	223,421	43.15%
Wednesday	254,275	45.05%
Thursday	194,258	43.57%
Friday	201,478	43.75%
Saturday	299,018	37.68%

Finding: Saturday has the highest session volume but a lower purchase rate (37.68%). Wednesday has the highest observed rate at 45.05%.

Actionable recommendation: Investigate why Saturday has high cart activity but lower conversion. Compare Saturday's category mix, session duration, cart-event depth and product mix with other days.

Overall SQL Takeaways

- **Abandonment is the central baseline:** 57.83% of sessions did not purchase.
- **Engagement is the strongest recurring pattern:** more distinct products, more cart events and longer sessions are all associated with higher purchase rates.
- **Single-product and single-event sessions deserve attention:** these are the largest groups and have much lower purchase rates than deeper sessions.
- **Category and brand matter:** purchase rates vary considerably across high-volume product segments.
- **Time patterns are secondary:** hour and day of week show variation, but the engagement-related differences are generally more pronounced.
- **Recommendations are hypotheses:** the SQL analysis is descriptive. Controlled experiments or richer event data would be needed to establish causal effects.

Data Quality and Analysis Notes

- The CSV contains only cart events; there is no separate purchase event type. The purchased boolean is used to identify whether a cart session ultimately purchased.
- Category coverage: 2,452,815 of 2,656,529 cart-event rows have a category; 137 unique categories were observed.
- Brand coverage: 2,399,890 of 2,656,529 cart-event rows have a brand; 3,383 unique brands were observed.
- Category and brand comparisons use a minimum threshold of 10,000 sessions to reduce instability from very small groups.
- Session duration is calculated from the first to last cart event in a session. It is not full website browsing duration.
- Time-of-day and day-of-week analysis uses the first cart event of each session.
- All conclusions are associations and should not be interpreted as causal effects.